Parametric Equations

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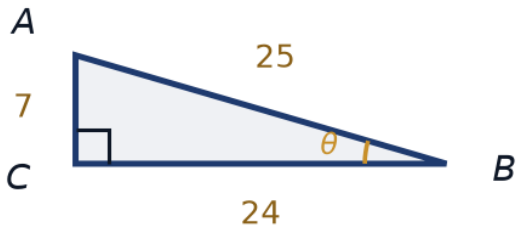
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Derivatives of parametric equations

For $x(t) = t^2$, $y(t) = t^3$, find $\frac{dy}{dx}$ in terms of t.

2 [Foundational] • Target: Derivatives of parametric equations

For $x(t) = \cos t$, $y(t) = \sin t$, find $\frac{dy}{dx}$ and evaluate it at $t = \frac{\pi}{4}$.



3 [Foundational] • Target: Tangent lines to parametric curves

Find the equation of the tangent line to the curve $x(t) = t^2 - 1$, $y(t) = t^3 - t$ at $t = 2$.

4 [Foundational] • Target: Arc length of parametric curves

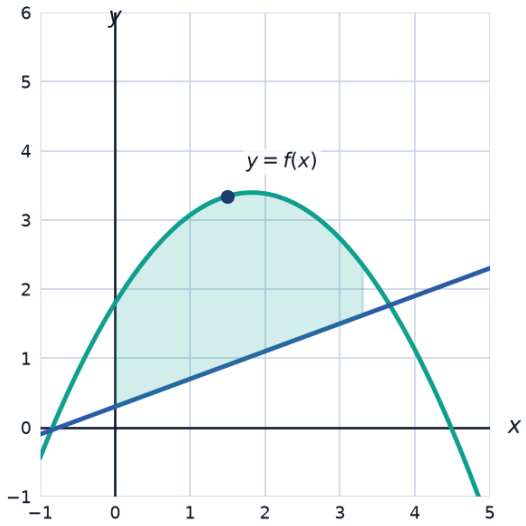
State the arc length formula for a parametric curve $x(t)$, $y(t)$ on $[a, b]$.

5 [Foundational] • Target: Arc length of parametric curves

Find the arc length of $x(t) = 3t$, $y(t) = 4t$ for $t \in [0, 2]$.

6 [Foundational] • Target: Arc length of parametric curves

Set up (but do not evaluate) the arc length integral for $x(t) = t^2$, $y(t) = t^3$ on $[0, 1]$.



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7 [Foundational] • Target: Speed as a vector quantity

A particle moves with position $x(t) = t^2$, $y(t) = 2t$. Find its speed at $t = 1$.

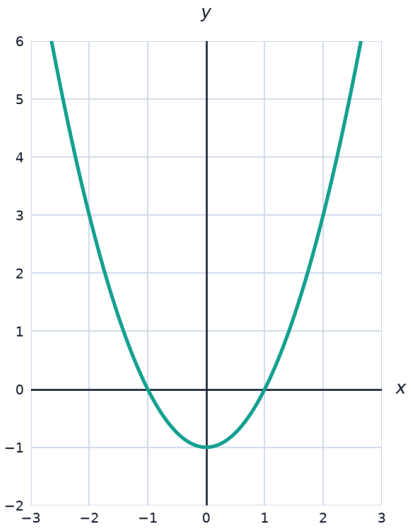
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Speed as a vector quantity

A particle moves with position $x(t) = 3\cos t$, $y(t) = 3\sin t$. Find its speed at any time t, and interpret the result.

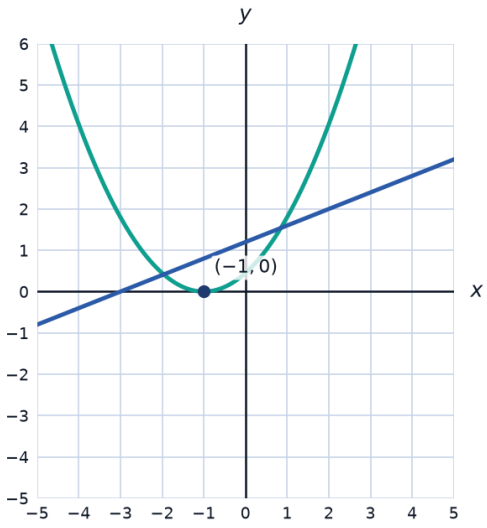
9 [Intermediate] • Target: Eliminating the parameter

Eliminate the parameter from $x(t) = t$, $y(t) = t^2 - 1$ to find the Cartesian equation, and sketch it.



10 [Intermediate] • Target: Eliminating the parameter

Eliminate the parameter from $x(t) = 2t + 1$, $y(t) = 4t - 3$ to find the Cartesian equation.



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11 [Intermediate] • Target: Eliminating the parameter

Explain why eliminating the parameter from $x(t) = \cos t$, $y(t) = \sin t$ using the identity $\sin^2 t + \cos^2 t = 1$ gives a relation that is NOT a function $y = f(x)$.

12 [Intermediate] • Target: Concavity of parametric curves

For $x(t) = t^2$, $y(t) = t^3$ (with $\frac{d^2y}{dx^2} = \frac{3}{4t}$ found earlier), determine the concavity of the curve for $t > 0$ and for $t < 0$.

13 [Intermediate] • Target: Horizontal and vertical tangents

For $x(t) = t^2 - 4$, $y(t) = t^3 - 3t$, find the value(s) of t where the tangent line is horizontal (where $\frac{dy}{dt} = 0$, provided $\frac{dx}{dt} \neq 0$).

14 [Intermediate] • Target: Horizontal and vertical tangents

For the same curve, find the value(s) of t where the tangent line is vertical (where $\frac{dx}{dt} = 0$, provided $\frac{dy}{dt} \neq 0$).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Motion along a parametric path (particle interpretation) • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

A particle's position is $x(t) = t^2$, $y(t) = t^3 - 3t$ for $t \in [-2, 2]$. Find all times when the particle is moving straight up or down (vertical velocity component only, i.e. $\frac{dx}{dt} = 0$).

16 [Challenging] • Target: Motion along a parametric path (particle interpretation) • Cross-Lesson: 6.01 Power Series and Interval of Convergence

For the particle above, find the speed at $t = 2$.



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17 [Challenging] • Target: Arc length of a full circle (verifying the circumference formula) • Cross-Lesson: 6.02 Taylor and Maclaurin Series

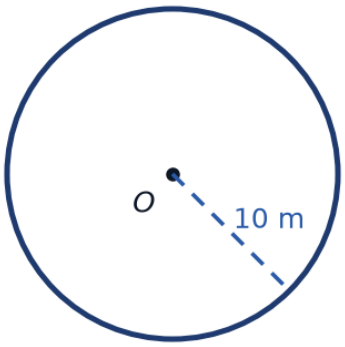
Use the parametric arc length formula to verify that the circumference of a circle $x(t) = r\cos t$, $y(t) = r\sin t$ for $t \in [0, 2\pi]$ is $2\pi r$.

18 [Challenging] • Target: Arc length of a full circle (verifying the circumference formula) • Cross-Lesson: 1.01 Integration by Parts

Find the arc length of the parametric curve $x(t) = e^t\cos t$, $y(t) = e^t\sin t$ is a spiral; set up (but do not evaluate) its arc length integral on $[0, \pi]$.

19 [Challenging] • Target: Parametric equations from real motion • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

A Ferris wheel of radius 10 m completes one revolution every 40 s, centered 12 m above the ground. Write parametric equations for a rider's height above the ground as a function of time t (seconds), assuming the rider starts at the bottom.



20 [Challenging] • Target: Parametric equations from real motion • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

For the Ferris wheel model above, find the rider's vertical velocity $\frac{dy}{dt}$ at $t = 10$ s (a quarter revolution).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Derivatives of parametric equations

Eliminate the parameter from $x(t) = -2t - 3$, $y(t) = 7t - 9$ to find the Cartesian equation, then state its slope.

a. $-\frac{7}{2}$

b. $-\frac{5}{2}$

c. $-\frac{2}{7}$

d. $\frac{7}{2}$

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22 [Foundational] • Target: Second derivatives of parametric equations

For $x(t) = t^2 + 2t$, $y(t) = t^3 + 6t^2$, find the larger value of t where the tangent line is horizontal.

a. -4

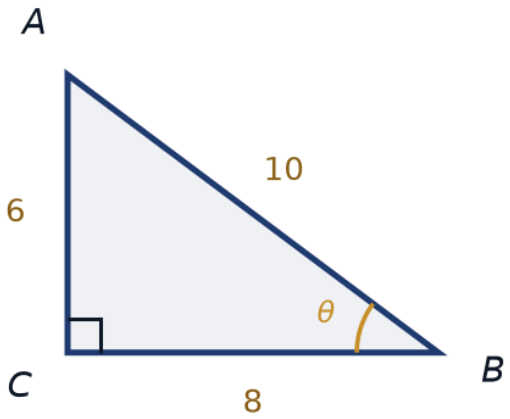
b. 0

c. -3

d. 3

23 [Foundational] • Target: Tangent lines to parametric curves

A particle's motion is described by the parametric equations $x(t) = e^{2t} - 2t$ and $y(t) = 4e^t$ for $t \geq 0$. Find the value of $\frac{dy}{dx}$ at the point where the tangent line to the curve is horizontal.



a. 0

b. 1

c. 2

d. 4

24 [Foundational] • Target: Arc length of parametric curves

A particle moves along $x(t) = 2t^2$, $y(t) = 1t^3$. Find its speed at $t = 2$.

a. $1 + 4\sqrt{13}$

b. $2\sqrt{13}$

c. $4\sqrt{13}$

d. 20

25 [Foundational] • Target: Speed as a vector quantity

A particle moves along parametric path $x(t) = 3t^2$, $y(t) = 5t$. What is the velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$?

a. $\langle 6t, 5 \rangle$

b. $\langle 3t, 5 \rangle$

c. $\langle 6t, 0 \rangle$

d. $\langle 3t^2, 5t \rangle$

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26 [Foundational] • Target: Eliminating the parameter

Find the arc length of $x = 3t^2,\ y = \frac{6}{3}t^3$ from $t = 0$ to $t = 2$.

a. $-1 + 5\sqrt{5}$

b. $-4 + 20\sqrt{5}$

c. $-2 + 10\sqrt{5}$

d. $-1 + 10\sqrt{5}$

27 [Foundational] • Target: Concavity of parametric curves

At a certain instant, a particle has velocity components $x'(t) = 3$ and $y'(t) = 4$. Find its speed.

a. $2\sqrt{3}$

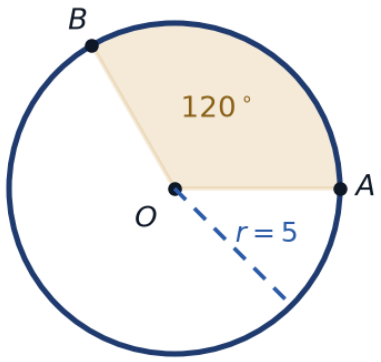
b. 5

c. 7

d. 25

28 [Foundational] • Target: Horizontal and vertical tangents

Arc length of parametric curve: $L = \int_a^b$



a. $\sqrt{(dx/dt) + (dy/dt)}\,dt$

b. $\sqrt{(dx/dt)^2 + (dy/dt)^2}\,dt$

c. $(dx/dt)^2 + (dy/dt)^2\,dt$

d. $|dy/dx|, dt$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Motion along a parametric path (particle interpretation)

A particle's position at time t is given by the vector-valued function $\mathbf{r}(t) = \langle 3t^2 + 5,\ -2t + 1\rangle$. Find the velocity vector $\mathbf{v}(t)$ at time t.

a. $\langle 3t^2 + 5,\ -2t + 1\rangle$

b. $\langle 6t,\ -2\rangle$

c. $\langle 3t,\ -2\rangle$

d. $\langle 6t^2,\ -2t\rangle$

30 [Intermediate] • Target: Arc length of a full circle (verifying the circumference formula)

A particle's position is described by the parametric equations $x(t) = 3t + 5$ and $y(t) = t^2 - 2$. What is the velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t)\rangle$ at time t?

a. $\langle 3, 2t\rangle$

b. $\langle 3, t^2 - 2\rangle$

c. $\langle 3t + 5, 2t\rangle$

d. $\langle 3, 2\rangle$

31 [Intermediate] • Target: Parametric equations from real motion

Consider a particle whose position is described by the parametric equations $x(t) = 3t + 5$ and $y(t) = 2t^2 - 1$ for $t \geq 0$. What is the velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t)\rangle$?

a. $\langle 3, 4t\rangle$

b. $\langle 3t, 4t\rangle$

c. $\langle 3, 2t\rangle$

d. $\langle 3, 4t - 1\rangle$

32 [Intermediate] • Target: Derivatives of parametric equations

A particle has velocity $\mathbf{r}'(t) = \langle 1,\ 3t\rangle$ and position $\mathbf{r}(0) = \langle 2,\ 1\rangle$. Find $\mathbf{r}(1)$.

a. $\langle 3,\ 5/2\rangle$

b. $\langle 2,\ 1\rangle$

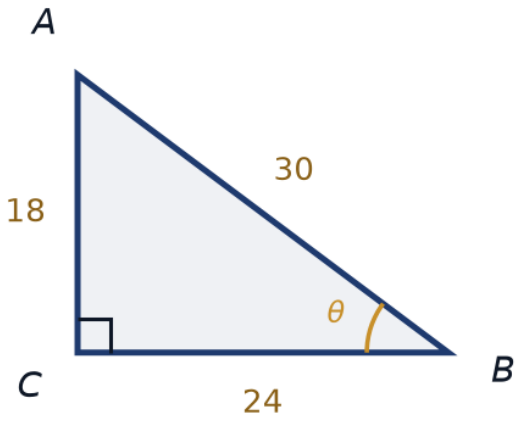
c. $\langle 5/2,\ 3\rangle$

d. $\langle 4,\ 5/2\rangle$

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33 [Intermediate] • Target: Second derivatives of parametric equations

A particle moves with $x(t) = 1\cos t,\ y(t) = 1\sin t$ from $t = 0$ to $t = 2\pi$. Find the total distance travelled.



a. 4π

b. π

c. 2π

d. 1

34 [Intermediate] • Target: Tangent lines to parametric curves

A particle moves with $x(t) = 3t^3,\ y(t) = 2t^2$. Find the acceleration vector at $t = 1$.

a. $\langle 18,\ 5\rangle$

b. $\langle 19,\ 4\rangle$

c. $\langle 4,\ 18\rangle$

d. $\langle 18,\ 4\rangle$

35 [Intermediate] • Target: Arc length of parametric curves

A particle moves with $x(t) = 1t^2,\ y(t) = 2t$. Find the velocity vector at $t = 1$.

a. $\langle 4,\ 2\rangle$

b. $\langle 3,\ 2\rangle$

c. $\langle 1,\ 2\rangle$

d. $\langle 2,\ 2\rangle$

36 [Intermediate] • Target: Speed as a vector quantity

A particle has velocity vector $\mathbf{r}'(t) = \langle 9,\ 12\rangle$ at a given instant. Compute $r'(t)$.

a. 21

b. 15

c. 225

d. $3\sqrt{7}$

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37 [Intermediate] • Target: Eliminating the parameter

Find the slope of the tangent line $\frac{dy}{dx}$ to the parametric curve $x(t) = t^2 + 1$, $y(t) = t^3 - 3t$ at $t = 2$.

a. $\frac{9}{4}$

b. $\frac{11}{4}$

c. $\frac{4}{9}$

d. $\frac{8}{4}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Concavity of parametric curves • Cross-Lesson: 1.01 Integration by Parts

A curve is given by $x(t) = t^2,\ y(t) = t^3 - 3t$. Find $\frac{dy}{dx}$ at $t = 1$.

a. 3

b. 0

c. 1

d. 2

39 [Challenging] • Target: Horizontal and vertical tangents • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

The cross product of two parallel vectors is:

a. A unit vector

b. The zero vector

c. Their dot product

d. Undefined

40 [Challenging] • Target: Motion along a parametric path (particle interpretation) • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Find the arc length of the spiral $r = 3\theta$ from $\theta = 0$ to $\theta = 2\pi$.

a. $3\text{asinh}(2\pi) + 6\pi\sqrt{1 + 4\pi^2}$

b. $\frac{3\text{asinh}(2\pi)}{2} + 3\pi\sqrt{1 + 4\pi^2}$

c. 6π

d. $\frac{3\text{asinh}(2\pi)}{4} + \frac{3\pi\sqrt{1 + 4\pi^2}}{2}$

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41 [Challenging] • Target: Arc length of a full circle (verifying the circumference formula) • Cross-Lesson: 3.02 Vector-Valued Functions

Find the total area enclosed by the polar cardioid $r = 2(1 + \cos\theta)$ for $0 \le \theta \le 2\pi$.

a. 4π

b. $\frac{12\pi}{2}$

c. $2 \cdot 4\pi$

d. $\frac{12\pi}{4}$

42 [Challenging] • Target: Parametric equations from real motion • Cross-Lesson: 4.01 Polar Curves and Derivatives

Consider a particle whose velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$ satisfies the differential equation $y'' - 2y' + y = 0$ and the initial conditions $x(0) = 1, y(0) = 2, x'(0) = 0, y'(0) = 1$. Furthermore, the particle's path is such that the magnitude of its acceleration vector is always equal to the magnitude of its velocity vector, i.e., $||\mathbf{a}(t)|| = ||\mathbf{v}(t)||$. Find the value of $x(1)$.

a. $e^{-1} + 1$

b. $2e^{-1} + 1$

c. $e^{-1} - 1$

d. $2 - e^{-1}$

43 [Challenging] • Target: Derivatives of parametric equations • Cross-Lesson: 4.02 Area in Polar Coordinates

A particle's position in the xy-plane is given by the parametric equations $x(t) = 2t + 1$ and $y(t) = t^2 - 3$, where t is time. What is the Cartesian equation of the path traced by the particle?

a. $y = \frac{1}{2}x^2 - 3$

b. $y = \left(\frac{x-1}{2}\right)^2 - 3$

c. $y = x^2 - 3$

d. $y = 2\left(\frac{x-1}{2}\right) + 1$

44 [Challenging] • Target: Second derivatives of parametric equations • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Epicycloid with n cusps (outer radius R, small circle radius $r = R/n$) has arc length:

a. $8nR$

b. $8R$

c. $4nR$

d. $2\pi nR$

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45 [Challenging] • Target: Tangent lines to parametric curves • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Arc length of $r = 2\cos\theta$ (full circle, $0 \le \theta \le \pi$):

a. π

b. 4

c. $\pi/2$

d. 2π

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